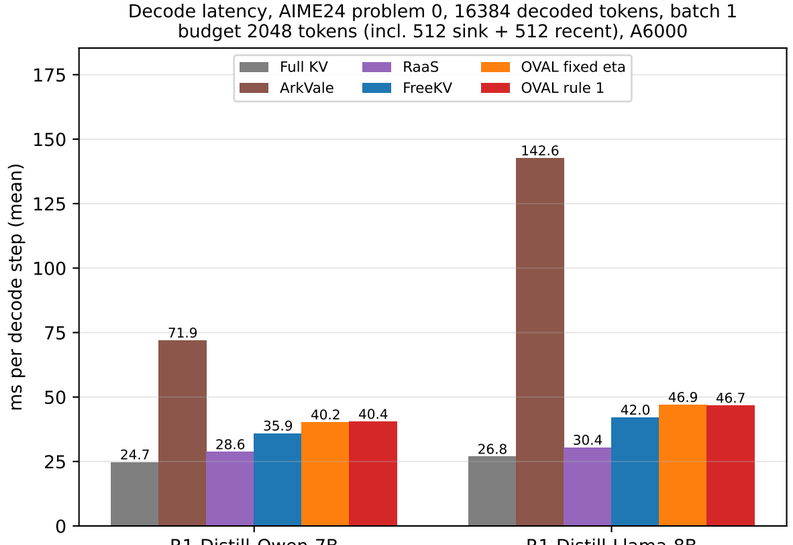
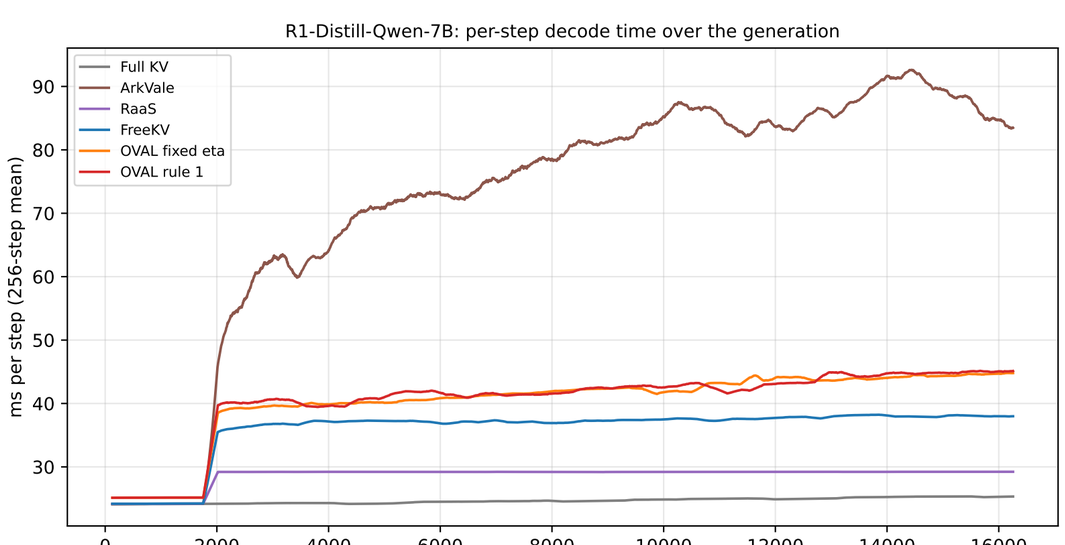
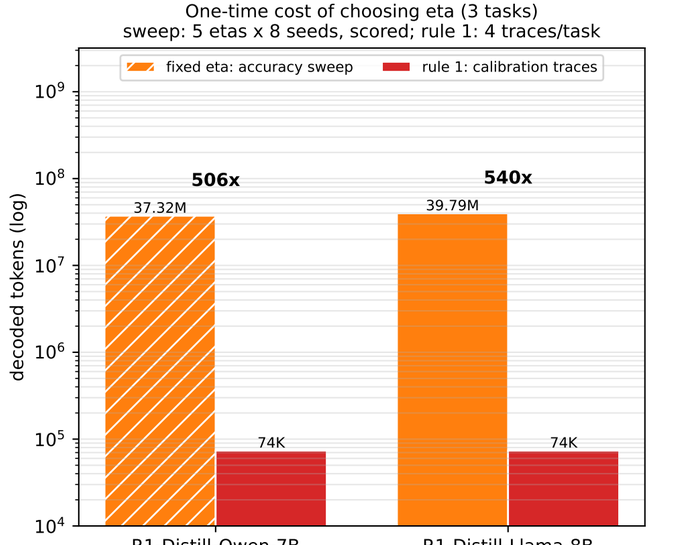

Decode latency, AIME24 problem 0, 16384 decoded tokens, batch 1
budget 2048 tokens (incl. 512 sink + 512 recent), A6000
Full KV
ArkVale
RaaS
FreeKV
OVAL fixed eta
OVAL rule 1
ms per decode step (mean)
0
25
50
75
100
125
150
175
24.7
71.9
28.6
35.9
40.2
40.4
26.8
142.6
30.4
42.0
46.9
46.7
R1-Distill-Qwen-7B
R1-Distill-Llama-8B


R1-Distill-Qwen-7B: per-step decode time over the generation
Full KV
ArkVale
RaaS
FreeKV
OVAL fixed eta
OVAL rule 1
ms per step (256-step mean)
30
40
50
60
70
80
90
0
2000
4000
6000
8000
10000
12000
14000
16000
decode step


One-time cost of choosing eta (3 tasks)
sweep: 5 etas x 8 seeds, scored; rule 1: 4 traces/task
fixed eta: accuracy sweep
rule 1: calibration traces
decoded tokens (log)
10^4
10^5
10^6
10^7
10^8
10^9
506x
37.32M
74K
540x
39.79M
74K
R1-Distill-Qwen-7B
(sweep est.)
R1-Distill-Llama-8B